

SyloPay Frontend UI Documentation

React Single Page Application Components & Storybook Guide

1. Architecture & Core Pages

The SyloPay frontend is built as a Single Page Application (SPA) using React, Vite, TypeScript, and TailwindCSS. It implements a smooth, multi-step Buy Now, Pay Later checkout pipeline:

Route	Page Name	Operational Purpose
/	CheckoutPage	Product catalogs displaying premium items (e.g., Samsung Galaxy S25 Ultra) and checkout initiation.
/quotation	QuotationPage	Visual selection of credit terms, number of installments (1-12), and DeFi APR comparison.
/contract	ContractPage	Personal customer information capture (Pix/PIX rampa) and Freighter Wallet signature.
/processing	ProcessingPage	On-chain registration feedback loop, Pix sandboxed QR Code generation, and webhook listening.
/dashboard	DashboardPage	Customer's portal to track and settle upcoming installments directly using USDC on-chain.

2. Global State Management (useBNPL)

A robust React Context hook (useBNPL) serves as the single source of truth, synchronizing checkout steps, product selections, user details, quote calculations, and Soroban contract states:

```
interface BNPLContextType {
  currentStep: 'checkout' | 'quotation' | 'contract' | 'processing' | 'dashboard';
  product: Product | null;
  selectedPlan: InstallmentPlan | null;
  personalInfo: PersonalInfo | null;
  contractId: string | null;
  walletAddress: string | null;
  setProduct: (p: Product) => void;
  selectPlan: (plan: InstallmentPlan) => void;
  savePersonalInfo: (info: PersonalInfo) => void;
  setContractId: (id: string) => void;
  setWalletAddress: (address: string) => void;
  resetCheckout: () => void;
}
```

3. Core Component Library

- **Button** — Robust button wrapping custom variations (primary, secondary, outline, ghost) and loading loaders.
- **ProgressBar** — Glowing orange checkout indicator tracking step checkout -> quotation -> contract -> processing -> dashboard.
- **Logo** — Displays official SyloPay typography and branding image.
- **WalletConnector** — Freightier wallet connector and details display.
- **PricingCalculator** — Compares BNPL checkout APR with credit card fees.
- **PixPayment** — Generates sandbox Pix copy-paste key and payment verification hooks.

4. UI Design System & Brand Aesthetics

SyloPay enforces a premium, dark-mode visual theme characterized by high-end typography and sleek shadows:

- **Primary Accent:** Glowing Amber Orange (#D97706) representing blockchain energy.
- **Dark Slate:** Off-black gradients (#0A0A0A, #121212) to create glassmorphism panels.
- **Text Hierarchy:** Off-white high contrast (#F4F4F5) and low contrast muted elements (#71717A).

5. Storybook Setup & Component Isolation

We have successfully integrated Storybook in the frontend workspace! Storybook isolated environments let developers build, refine, and unit test UI components without running the backend.

To run the searchable, interactive Storybook dashboard local server:

```
cd frontend
npm run storybook
```